SILVER OAK UNIVERSITY
EDUCATION TO INNOVATION

Minor Project
1010073491

Smart Wireless Honeypot and Real-Time Security Operations Center (SOC) Dashboard

Project ID	CSE_CS_07
Enrollment No	2401030730017
Name	Jaimin Makwana
Branch	CSE_CS
Guide Name	Prof. Harshita Makwana

Index

- Introduction
- Background and Motivation
- Relevance and Importance
- Supported Threat Vectors & Attack Surface
- Literature review
- Key Objectives
- System Architecture & Data Flow
- System Tech Stack & Tools
- System Implementation & Output Screenshots
- Method / Approach

Introduction

The Smart Wireless Honeypot and Real Time Security Operations Center (SOC) Dashboard is a cyber security project designed to improve wireless network security. It creates a fake wireless network that attracts suspicious users or attackers and safely records their activities. The collected information is displayed on a real time SOC Dashboard, where the administrator can monitor network events, detect suspicious behavior, and analyze security threats. This project helps in understanding wireless security, threat monitoring, and security operations in a practical and effective way.

This project belongs to the field of **Cyber Security**, with a focus on **Wireless Network Security** and **Security Operations Center (SOC)**. The research area aims to improve the detection and monitoring of suspicious activities in wireless networks using honeypot technology. The project develops a **Smart Wireless Honeypot** that attracts potential attackers in a safe environment and records their activities. The collected information is displayed on a **Real Time SOC Dashboard**, which helps administrators monitor security events, analyze threats, and improve overall network security.

Background and Motivation

Background

- Wi-Fi networks (802.11) are highly vulnerable to Layer-2 Management Frame attacks (Deauth, Beacon/Probe Floods).
- Traditional SIEMs and Network Firewalls only monitor Layer-3/4 wired traffic and miss RF-level wireless exploits.
- Enterprise Wireless Intrusion Detection Systems (WIDS) are cost-prohibitive for small setups and research labs.

Motivation

- To bridge the gap by deploying a low-cost, hardware-integrated Wireless Honeypot & Sniffer using ESP32 and ESP8266.
- To capture and detect multiple wireless attack vectors simultaneously (Deauth, Beacon, and Probe floods) using custom burst thresholds.
- To bridge hardware RF packet sniffing with a modern web dashboard via serial communication and WebSockets for instant SOC alert visualization.

Relevance and Importance

Addressing Critical Layer-2 Wi-Fi Vulnerabilities

- Modern security systems focuses mainly on Layer-3/4 network traffic, leaving Layer-2 Wi-Fi management frames unmonitored.
- Our system directly targets this gap by capturing unencrypted management frame exploits like Deauth Floods, Beacon Floods, and Probe Request Floods in real-time.

Cost-Effective Hardware-Based WIDS

- Replaces expensive enterprise Wireless Intrusion Detection Systems (WIDS) with low-cost, high-performance hardware (ESP32 Sniffer & ESP8266 Attacker/Honeypot).
- Makes enterprise-grade Wi-Fi threat detection accessible for educational institutions, research labs, and small-to-medium businesses.

Real-Time Threat Visualization for SOC Operations

- Bridges physical RF packet capturing with software-defined threat intelligence via Node.js and WebSockets.
- Converts complex wireless raw logs into actionable, live security alerts on the HoneyGrid SOC Dashboard, enabling immediate incident response.

Supported Threat Vectors & Attack Surface

1. Deauth Flood (Deauthentication Attack)
- **Category:** Wireless Layer-2 Management Frame Attack
- **Working:** The attacker transmits spoofed 802.11 deauthentication management frames (0x0C) between the target client and the Access Point to forcibly sever their wireless connection.
- **Project Role:** Forces target devices off legitimate networks so they automatically attempt reconnection—redirecting them toward our active Honeypot AP. The ESP32 hardware sniffer captures these frames using custom burst-threshold detection.

2. Beacon Flood
- **Category:** Wireless Layer-2 SSID Spamming
- **Working:** The attacker broadcasts thousands of fake Access Point beacons simultaneously with randomly generated SSIDs across multiple channels.
- **Project Role:** Spams and overloads nearby Wi-Fi network scanners and client devices. The ESP32 sniffer identifies high-frequency beacon frame spikes and pushes real-time alerts to the Honeyport SOC dashboard.

3. Probe Flood (Probe Request Flood)

- **Category:** Wireless Reconnaissance & Scanning
- **Working:** The attacker continuously floods the RF spectrum with automated probe requests (0x04) to discover active Access Points and gather network signatures.
- **Project Role:** Detects aggressive network scanning and client tracking activity within a strict 1.2-second evaluation window using the ESP32 promiscuous mode engine.

4. Evil Twin AP

- **Category:** Rogue Access Point / Wireless Spoofing
- **Working:** The attacker deploys a rogue Access Point that clones the exact SSID, MAC address (BSSID), and channel of a legitimate target Wi-Fi network.
- **Project Role:** Tricks nearby client devices into authenticating with the Honeypot infrastructure, enabling transparent monitoring of unencrypted traffic and capture of login credentials.

5. Karma Attack

- **Category:** Automatic AP Spoofing / Decoy Mechanism
- **Working:** Listens for broadcast probe requests emitted by nearby devices querying their Preferred Network List (PNL) and dynamically responds claiming to be that specific network.
- **Project Role:** Seamlessly connects unauthenticated target devices to the Honeypot AP without requiring passwords, giving the SOC immediate visibility into client behaviour.

6. Brute Force
- **Category:** Authentication / Credential Exploitation
- **Working:** Automated dictionary attacks continuously attempt high-velocity username/password combinations against authentication services (HTTP/SSH/Captive Portal).
- **Project Role:** Monitors failed authentication limits on trapped Honeypot sessions and logs suspicious credential guessing attempts directly to the SOC dashboard.

7. DoS / HTTP Flood
- **Category:** Layer-7 Application Layer Attack
- **Working:** Floods the Honeypot web server and captive portal with an overwhelming volume of simultaneous HTTP GET/POST requests to exhaust CPU and RAM.
- **Project Role:** Tracks incoming web application request rates and triggers high-severity denial-of-service alerts when bandwidth or request thresholds are exceeded.

8. Port Scan & SSH
- **Category:** Network Reconnaissance & Exploitation
- **Working:** Attackers execute port scanning tools (e.g., Nmap) to discover open services (Port 22/SSH, 80/HTTP, 443/HTTPS) and attempt remote shell exploitation.
- **Project Role:** Captures unauthorized port probing and flags unauthorized SSH connection attempts aimed at Honeypot administrative ports.

9. ARP / MITM (Address Resolution Protocol / Man-In-The-Middle)
- **Category:** Layer-2 Network Traffic Interception
- **Working:** Transmits malicious ARP responses (ARP Cache Poisoning) across the local network to corrupt IP-to-MAC address mappings between clients and the gateway.
- **Project Role:** Detects rogue ARP frame broadcasts within the Honeypot subnet to prevent or report illegal packet interception and payload tampering in real-time.

Literature Review

Key Domains in Prior Wireless Security Research

802.11 Layer-2 MGMT Attacks

Deauth, Beacon, and Probe flooding research focuses heavily on frame analysis and RSSI anomaly detection.

Rogue AP & Decoy Systems

Studies on Evil Twin and Karma frameworks explore auto-responder mechanisms and PNL exploitation techniques

Network & Layer-7 Attacks

Literature covers SSH brute force, HTTP GET floods, port scans, and local ARP cache poisoning models.

Limitations in Existing Wireless Honeypots

Isolated Detection Engines
Existing tools detect wireless management floods or network layer attacks separately, lacking unified multi-layer correlation.

Lack of Real-Time SOC Integration
Most studies utilize static post-capture pcap analysis rather than pushing live telemetry to a centralized SOC dashboard.

High Cost & Power Demands
Traditional solutions depend on expensive enterprise Linux servers rather than low-cost, portable microcontrollers like ESP32.

Bridging Research Gaps in Honeyport

Multi-Layer Defense

Combines 802.11 Layer-2 frame sniffing with Layer-7 honeypot trapping to monitor all 9 specific threat vectors simultaneously.

Real-Time SOC Visualization

Eliminates manual pcap inspections by pushing live telemetry alerts to a unified web dashboard instantly.

Key Objectives

Wireless Honeypot Deployment
Set up a dedicated Wi-Fi target node to capture unauthorized probe requests and connection attempts safely.

Promiscuous Packet Capture
Program ESP32 microcontroller with channel hopping (1-13) to capture 802.11 management frames in real time.

Real-Time SOC Dashboard
Develop a web dashboard using Node.js and Express to display incoming security alerts with severity indicators.

Multi-Attack Simulation Engine
Create automated testing scripts to simulate Brute Force, Port Scans, and ARP Spoofing alongside physical RF attacks.

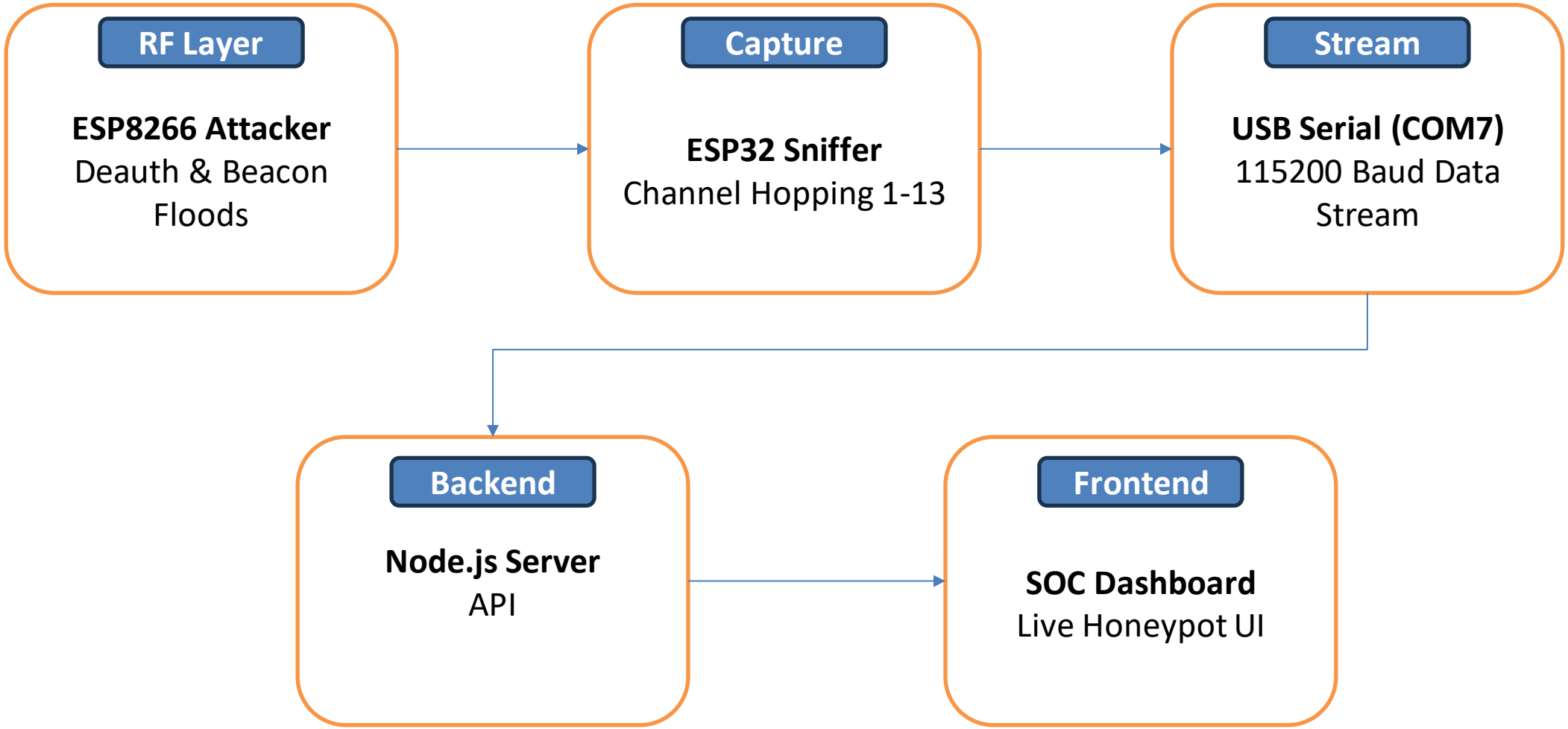
System Tech Stack & Tools

Hardware
- ✓ ESP32 Microcontroller: Wi-Fi promiscuous sniffer
- ✓ ESP8266 Board: Threat payload generator
- ✓ USB Serial Interface: COM7 connection at 115200 baud

Software & Backend
- ✓ Node.js & Express: Web server and REST API
- ✓ Serial Port Node Library: Hardware stream reader
- ✓ HTML5/ CSS3 / JavaScript: Dashboard UI frontend

Testing & Analysis
- ✓ Wireshark: 802.11 frame validation
- ✓ Arduino IDE: C++ firmware
- ✓ development Spacehuhn Firmware: ESP8266 Deauther platform

System Architecture & Data Flow



System Architecture & Data Flow

Hardware Path:
Physical 802.11 RF Management Frames → ESP32 Sniffer → Serial Buffer → Dashboard UI

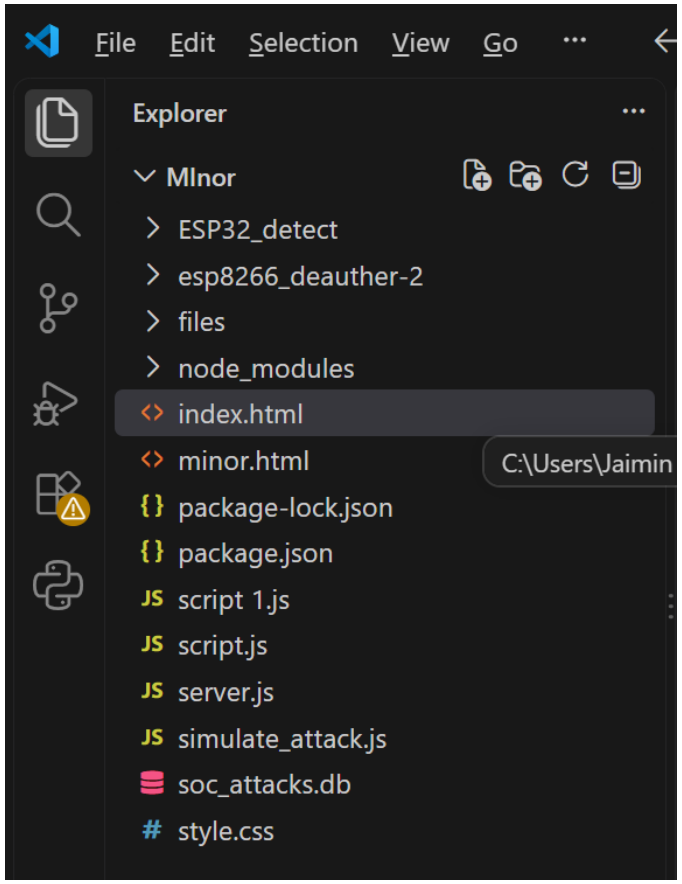
Software Simulation Path:
simulate_attack.js → HTTP POST (/api/simulate) → Node Server → Dashboard UI

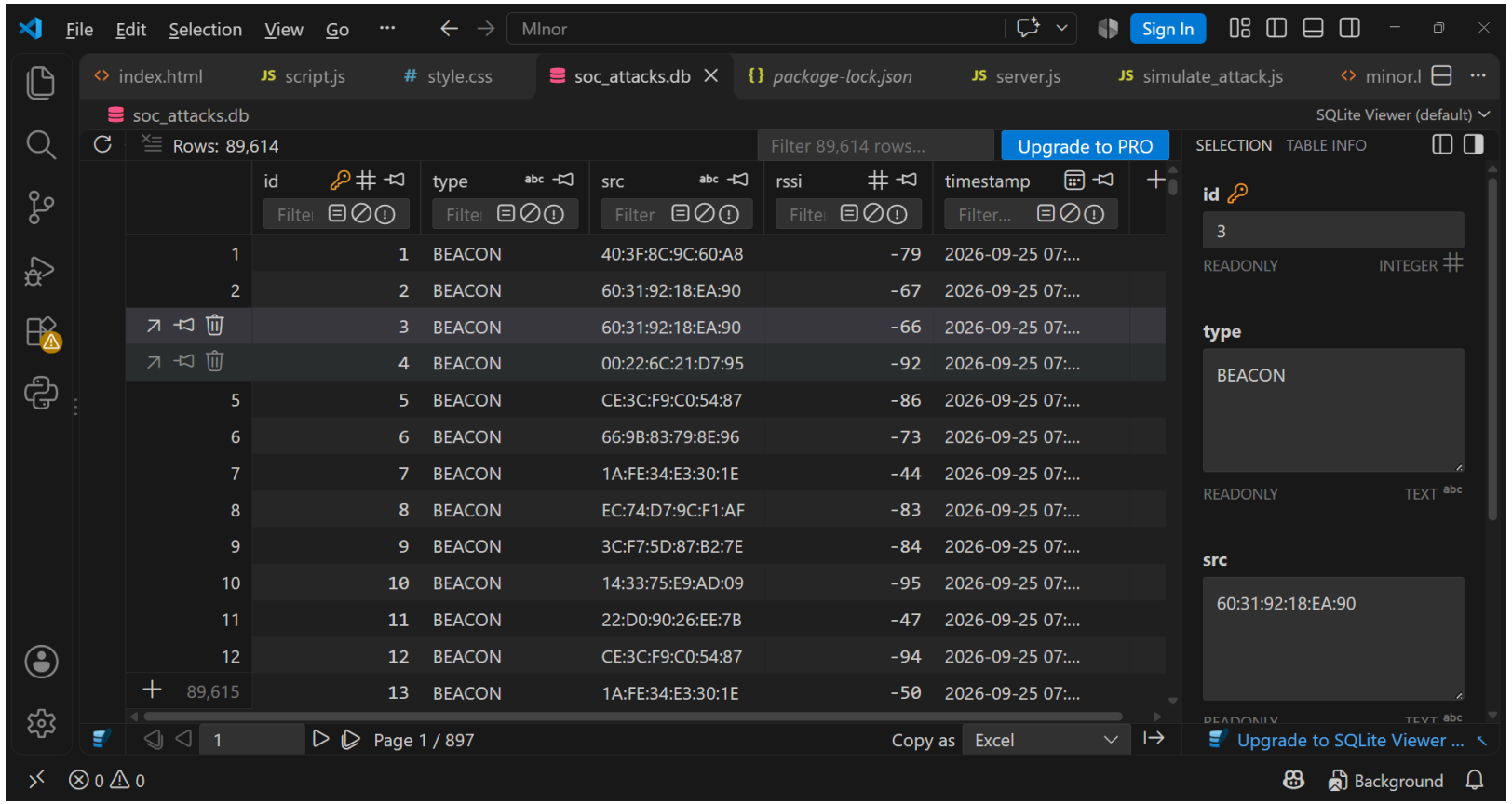
Method / Approach

- **Modular Directory Structure:** Organized file layout separating backend server logic, frontend dashboard UI, and microcontroller firmware.
- **Firmware Integration:** Separate scripts for ESP32 promiscuous packet sniffing and ESP8266 attack simulation nodes.
- **Real-time:** Establishes seamless serial communication between hardware sensors and the Node.js/Web socket backend.

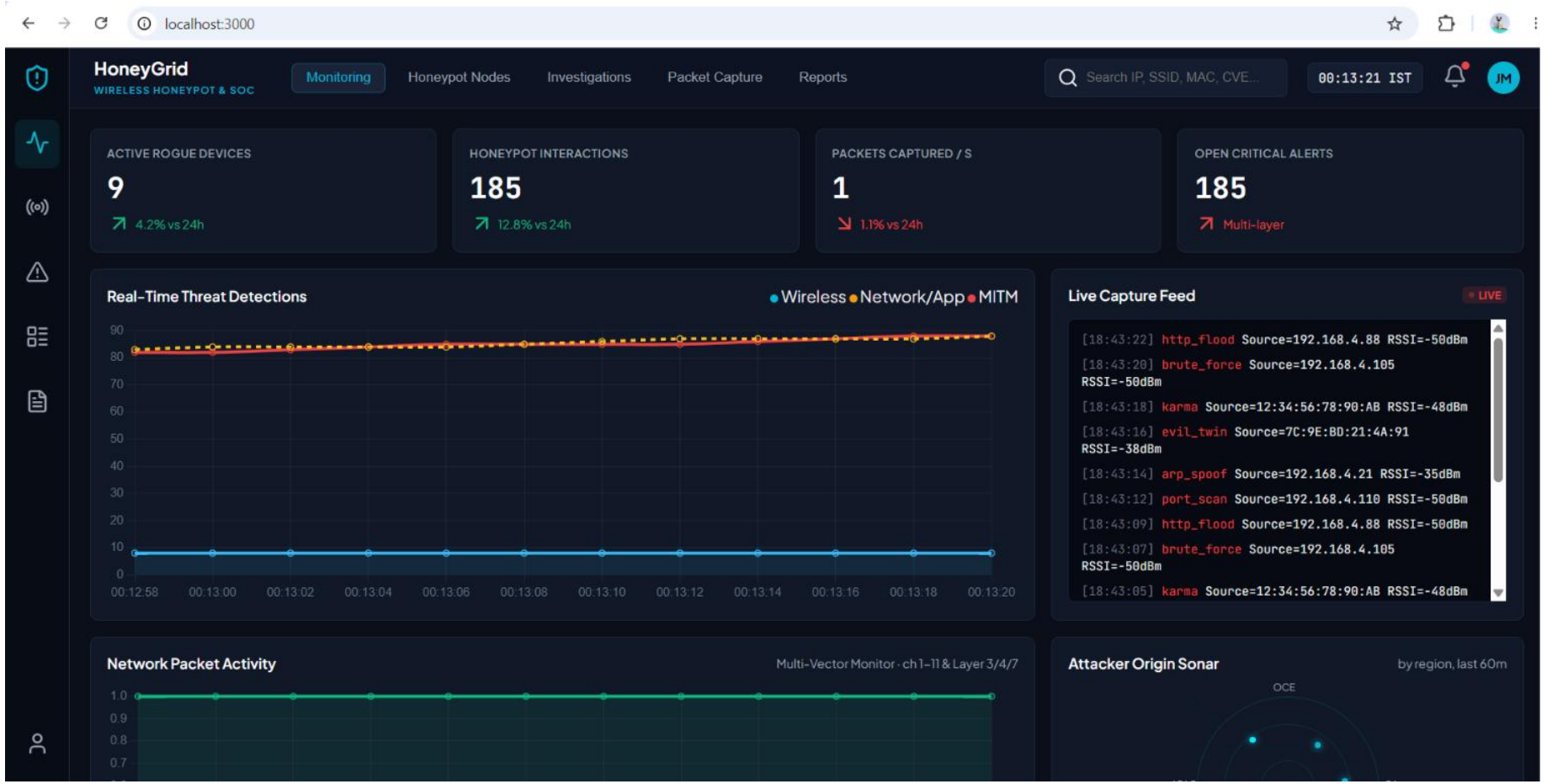
System Implementation & Output Screenshots

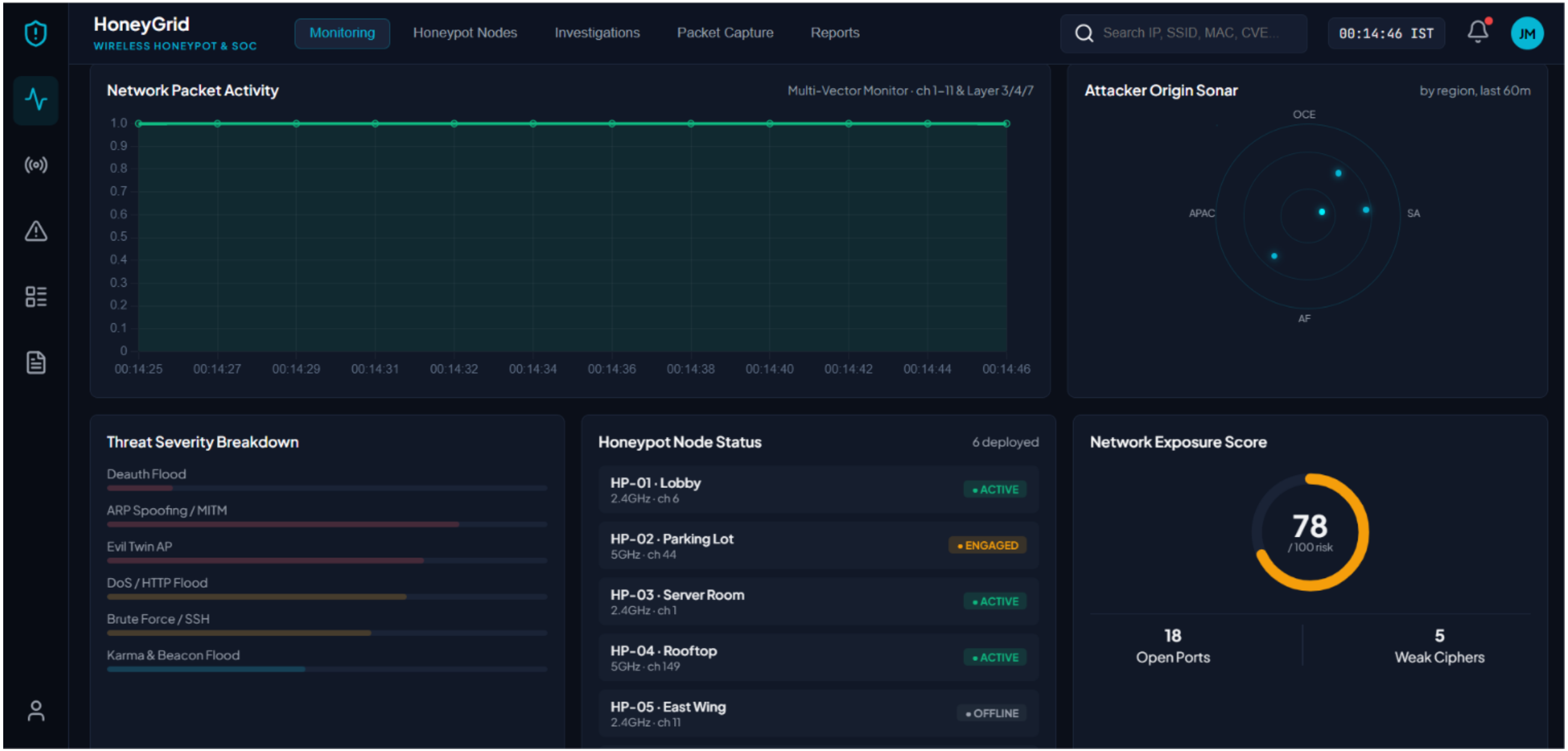
Project Directory & Workspace File Structure

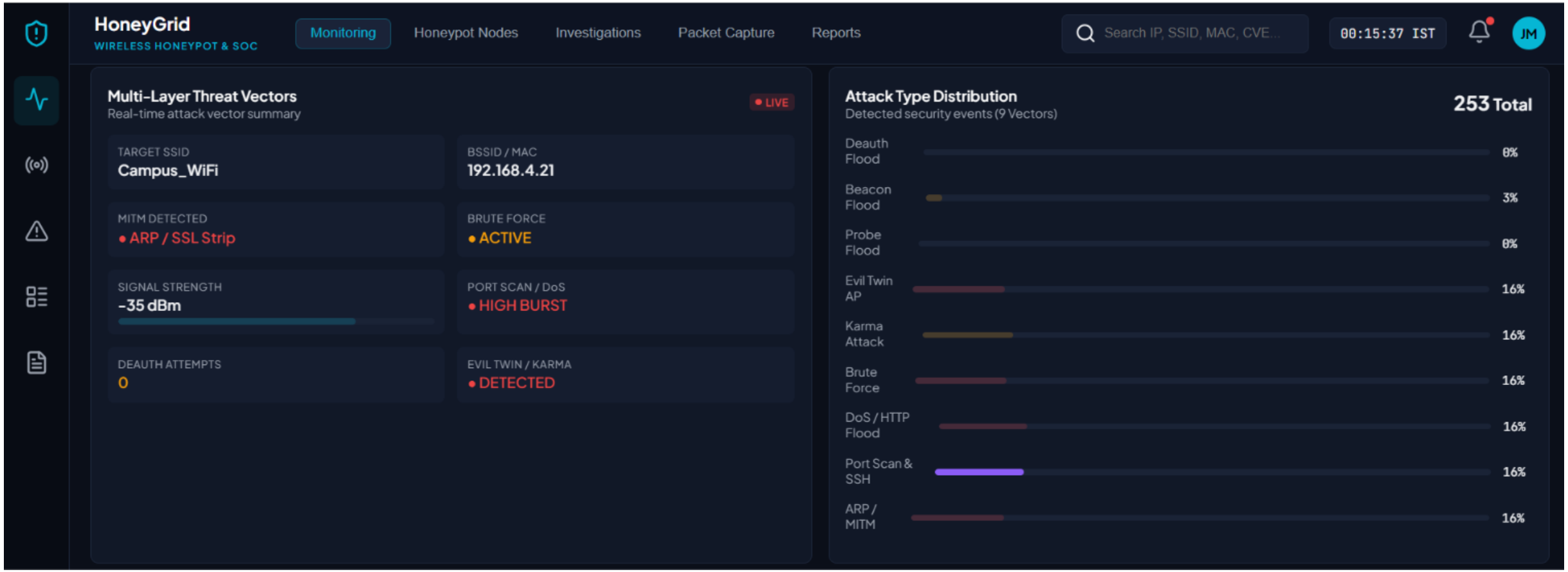


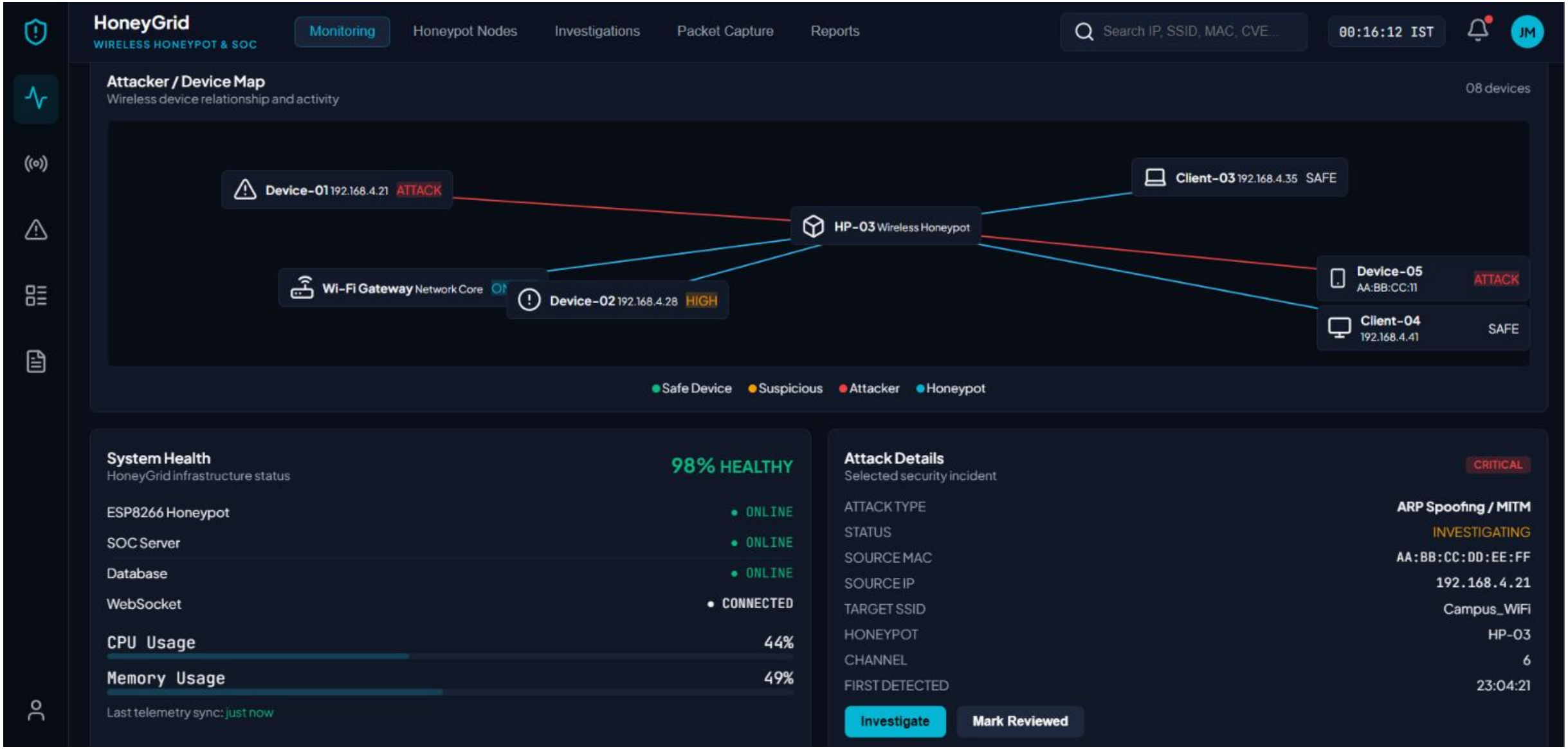


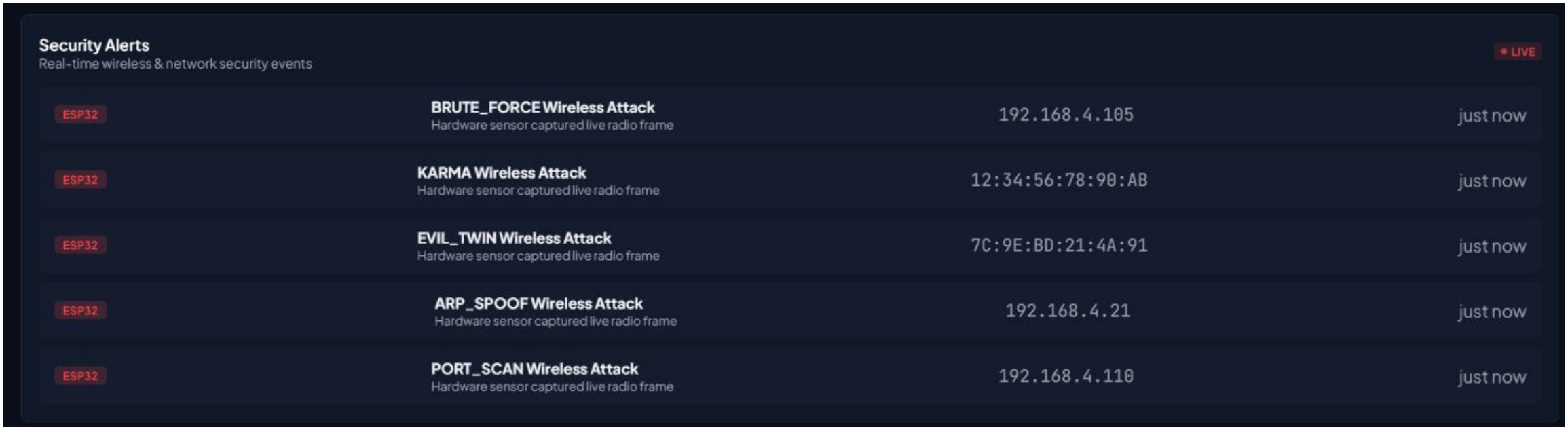
Real-Time Honeypot SOC Dashboard



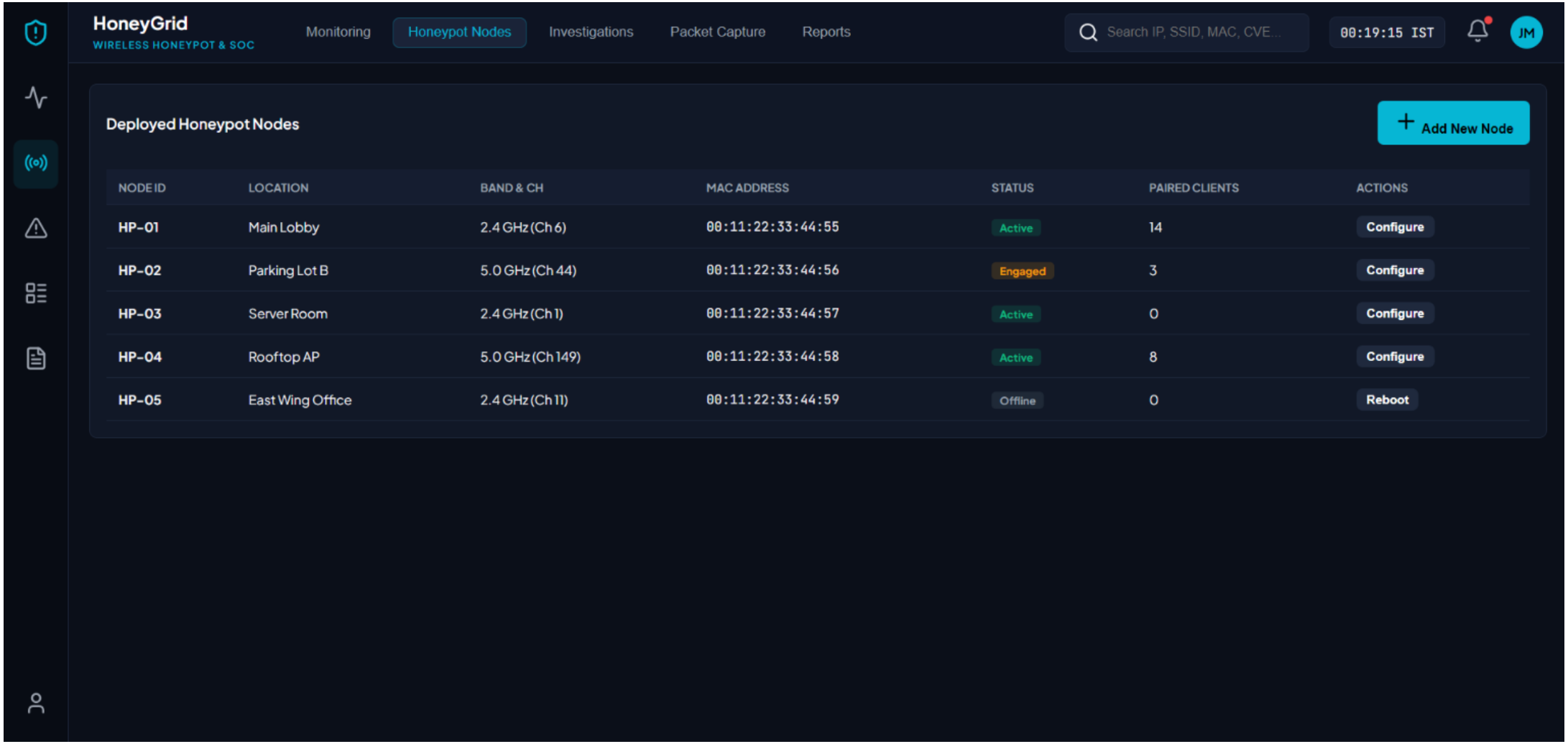

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:14:46 IST
JM
Network Packet Activity
Multi-Vector Monitor · ch 1–11 & Layer 3/4/7
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
00:14:25
00:14:27
00:14:29
00:14:31
00:14:32
00:14:34
00:14:36
00:14:38
00:14:40
00:14:42
00:14:44
00:14:46
Attacker Origin Sonar
by region, last 60m
OCE
APAC
SA
AF
Threat Severity Breakdown
Deauth Flood
ARP Spoofing / MITM
Evil Twin AP
DoS / HTTP Flood
Brute Force / SSH
Karma & Beacon Flood
Honeypot Node Status
6 deployed
HP-01 · Lobby
2.4GHz · ch 6
ACTIVE
HP-02 · Parking Lot
5GHz · ch 44
ENGAGED
HP-03 · Server Room
2.4GHz · ch 1
ACTIVE
HP-04 · Rooftop
5GHz · ch 149
ACTIVE
HP-05 · East Wing
2.4GHz · ch 11
OFFLINE
Network Exposure Score
78
/ 100 risk
18
Open Ports
5
Weak Ciphers

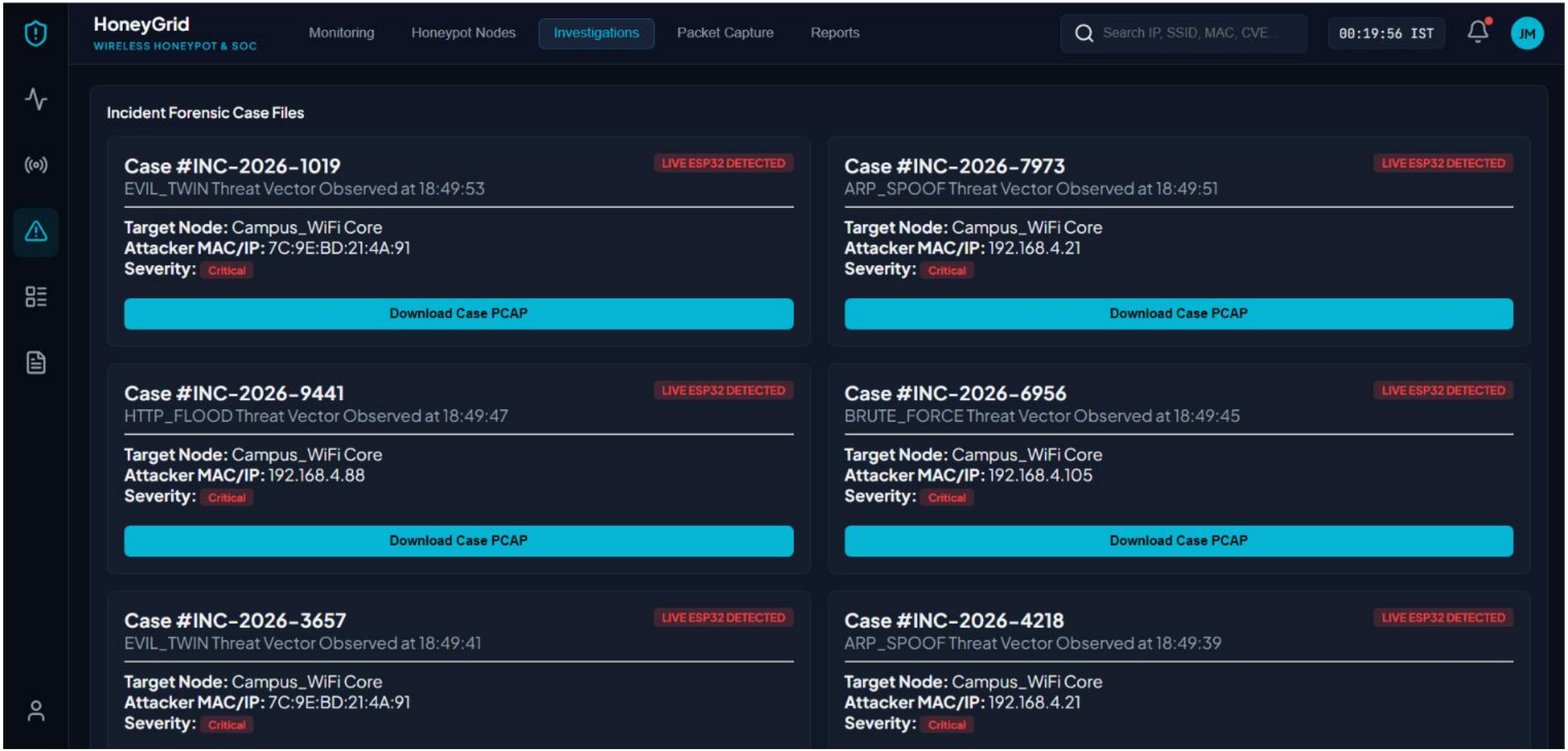

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:15:37 IST
JM
Multi-Layer Threat Vectors
Real-time attack vector summary
LIVE
TARGET SSID
Campus_WiFi
BSSID / MAC
192.168.4.21
MITM DETECTED
ARP / SSL Strip
BRUTE FORCE
ACTIVE
SIGNAL STRENGTH
-35 dBm
PORT SCAN / DoS
HIGH BURST
DEAUTH ATTEMPTS
0
EVIL TWIN / KARMA
DETECTED
Attack Type Distribution
Detected security events (9 Vectors)
253 Total
Deauth Flood
0%
Beacon Flood
3%
Probe Flood
0%
Evil Twin AP
16%
Karma Attack
16%
Brute Force
16%
DoS / HTTP Flood
16%
Port Scan & SSH
16%
ARP / MITM
16%

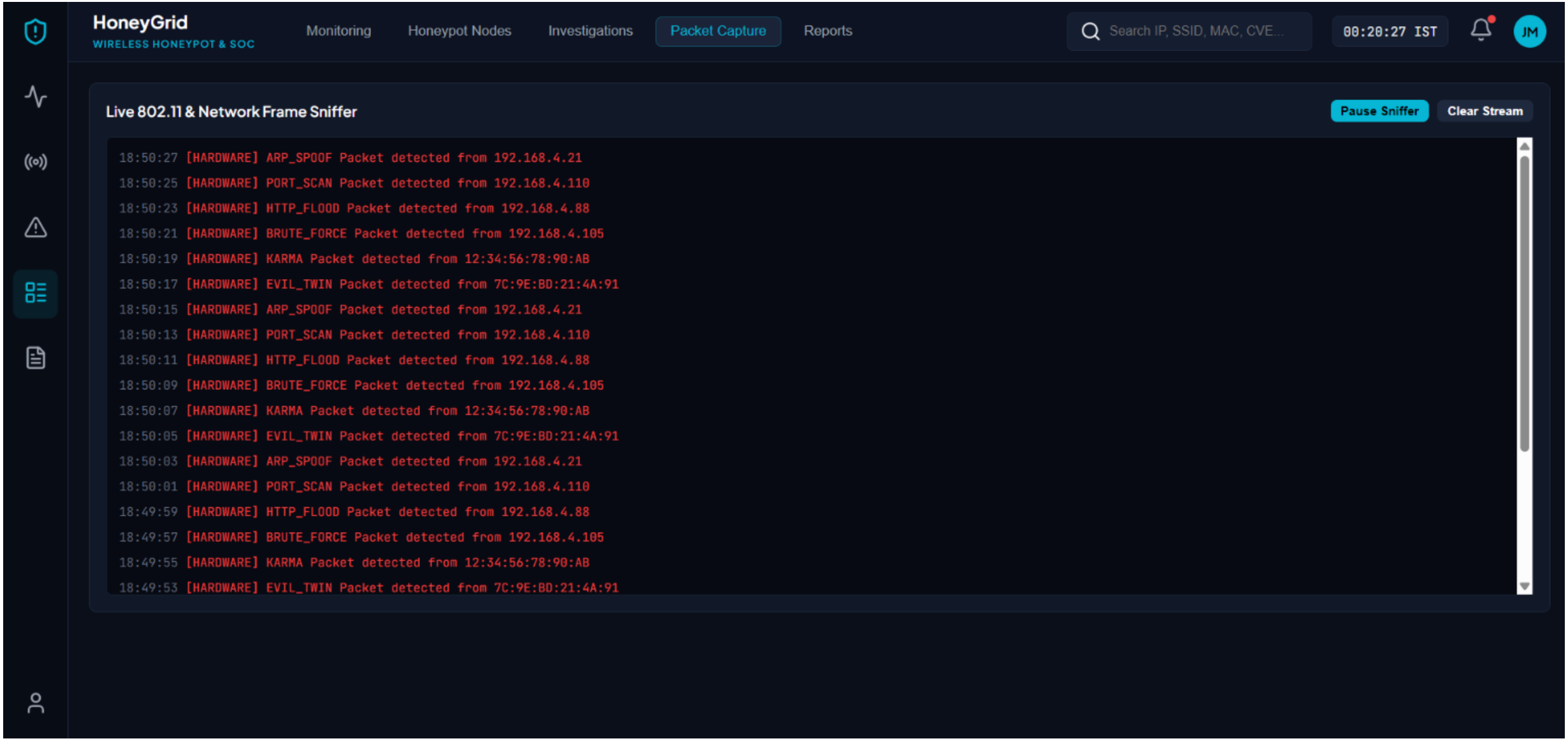

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:16:12 IST
JM
Attacker / Device Map
Wireless device relationship and activity
08 devices
Device-01 192.168.4.21 ATTACK
HP-03 Wireless Honeypot
Client-03 192.168.4.35 SAFE
Wi-Fi Gateway Network Core
Device-02 192.168.4.28 HIGH
Device-05 AA:BB:CC:11 ATTACK
Client-04 192.168.4.41 SAFE
Safe Device
Suspicious
Attacker
Honeypot
System Health
HoneyGrid infrastructure status
98% HEALTHY
ESP8266 Honeypot ONLINE
SOC Server ONLINE
Database ONLINE
WebSocket CONNECTED
CPU Usage 44%
Memory Usage 49%
Last telemetry sync: just now
Attack Details
Selected security incident
CRITICAL
ATTACK TYPE ARP Spoofing / MITM
STATUS INVESTIGATING
SOURCE MAC AA:BB:CC:DD:EE:FF
SOURCE IP 192.168.4.21
TARGET SSID Campus_WiFi
HONEYPOT HP-03
CHANNEL 6
FIRST DETECTED 23:04:21
Investigate
Mark Reviewed

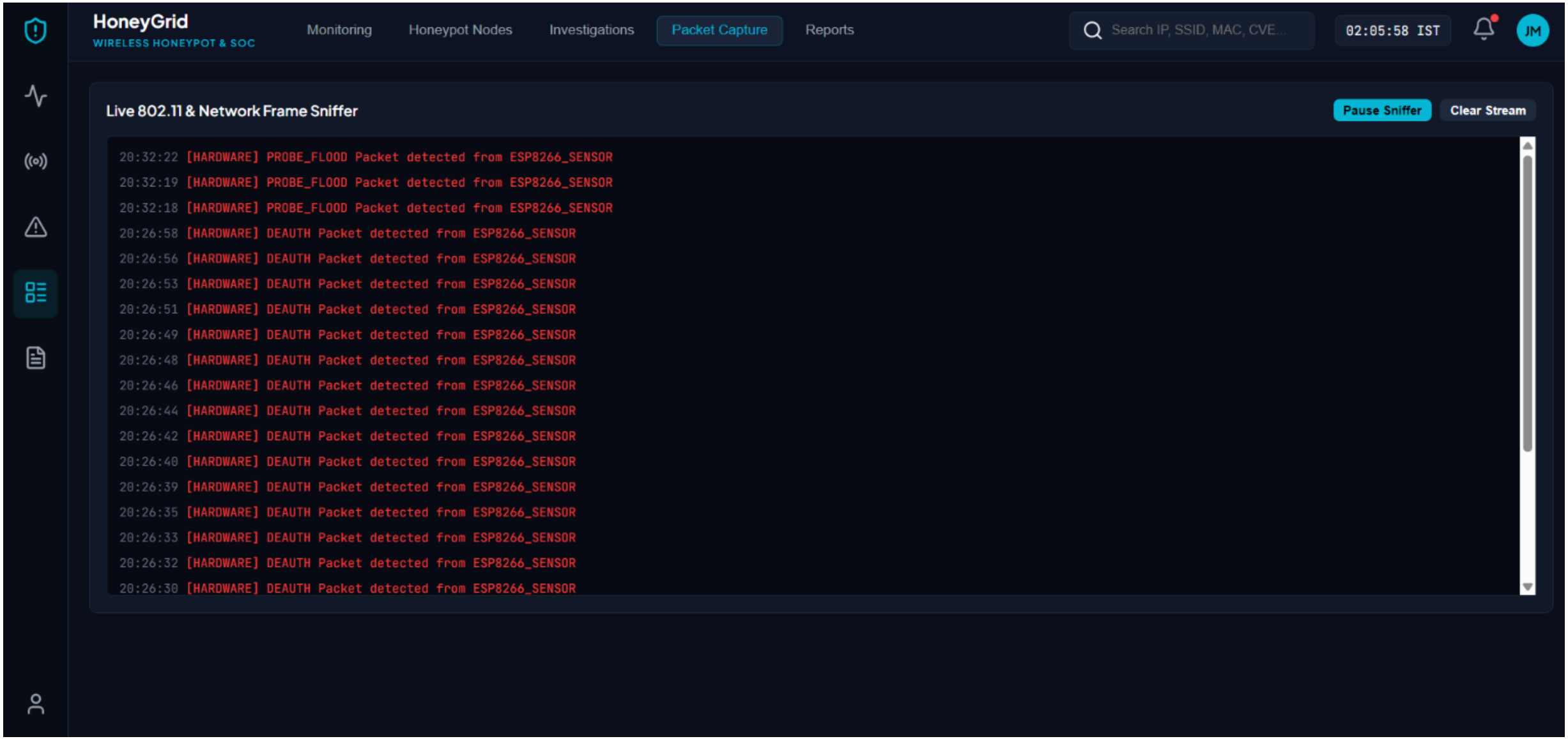

Security Alerts
Real-time wireless & network security events
LIVE
ESP32
BRUTE_FORCE Wireless Attack
Hardware sensor captured live radio frame
192.168.4.105
just now
ESP32
KARMA Wireless Attack
Hardware sensor captured live radio frame
12:34:56:78:90:AB
just now
ESP32
EVIL_TWIN Wireless Attack
Hardware sensor captured live radio frame
7C:9E:BD:21:4A:91
just now
ESP32
ARP_SPOOF Wireless Attack
Hardware sensor captured live radio frame
192.168.4.21
just now
ESP32
PORT_SCAN Wireless Attack
Hardware sensor captured live radio frame
192.168.4.110
just now

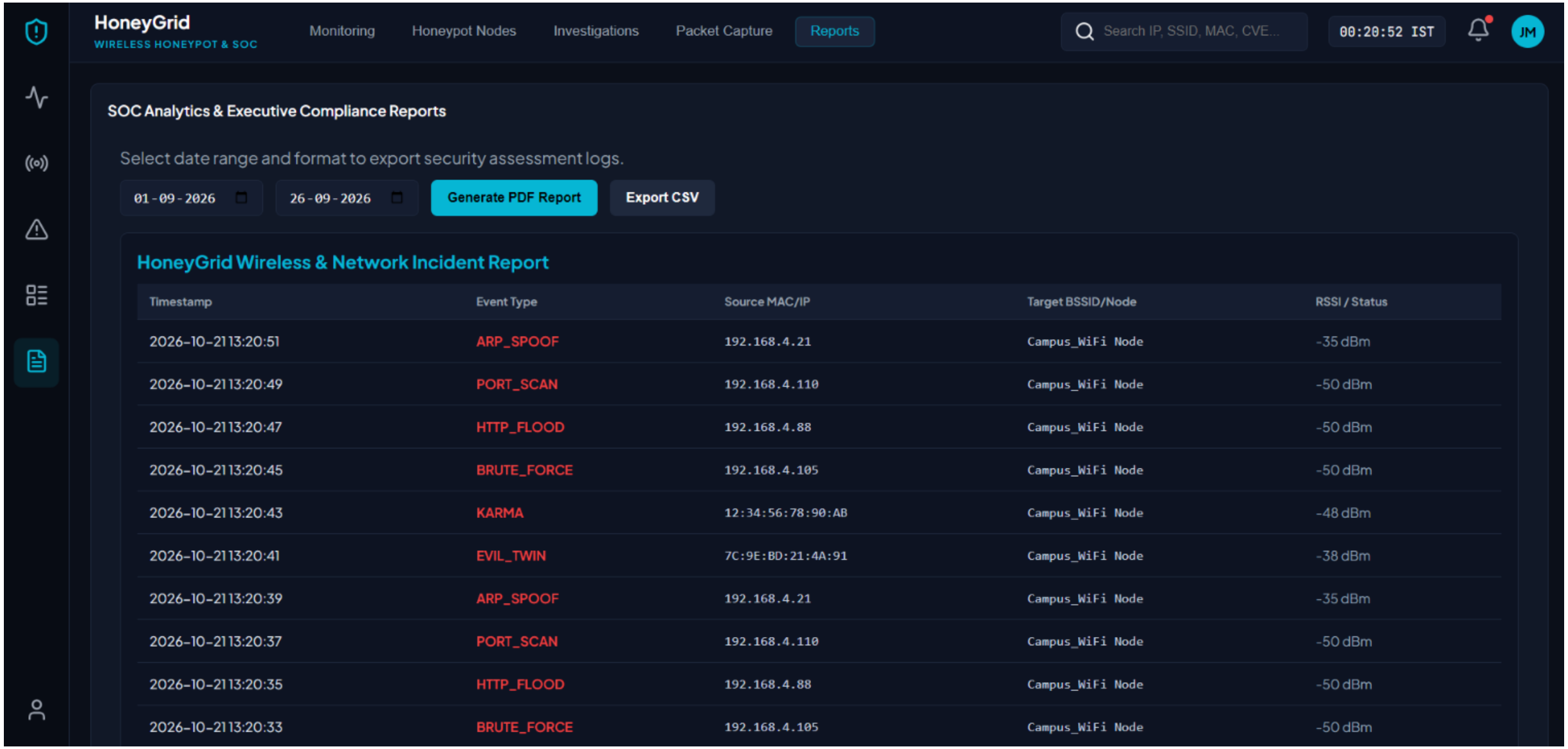
Live Attack Alert & Threat Monitoring



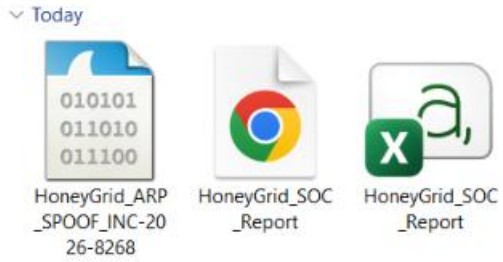

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:19:56 IST
JM
Incident Forensic Case Files
Case #INC-2026-1019
LIVE ESP32 DETECTED
EVIL_TWIN Threat Vector Observed at 18:49:53
Target Node: Campus_WiFi Core
Attacker MAC/IP: 7C:9E:BD:21:4A:91
Severity: Critical
Download Case PCAP
Case #INC-2026-7973
LIVE ESP32 DETECTED
ARP_SPOOF Threat Vector Observed at 18:49:51
Target Node: Campus_WiFi Core
Attacker MAC/IP: 192.168.4.21
Severity: Critical
Download Case PCAP
Case #INC-2026-9441
LIVE ESP32 DETECTED
HTTP_FLOOD Threat Vector Observed at 18:49:47
Target Node: Campus_WiFi Core
Attacker MAC/IP: 192.168.4.88
Severity: Critical
Download Case PCAP
Case #INC-2026-6956
LIVE ESP32 DETECTED
BRUTE_FORCE Threat Vector Observed at 18:49:45
Target Node: Campus_WiFi Core
Attacker MAC/IP: 192.168.4.105
Severity: Critical
Download Case PCAP
Case #INC-2026-3657
LIVE ESP32 DETECTED
EVIL_TWIN Threat Vector Observed at 18:49:41
Target Node: Campus_WiFi Core
Attacker MAC/IP: 7C:9E:BD:21:4A:91
Severity: Critical
Case #INC-2026-4218
LIVE ESP32 DETECTED
ARP_SPOOF Threat Vector Observed at 18:49:39
Target Node: Campus_WiFi Core
Attacker MAC/IP: 192.168.4.21
Severity: Critical

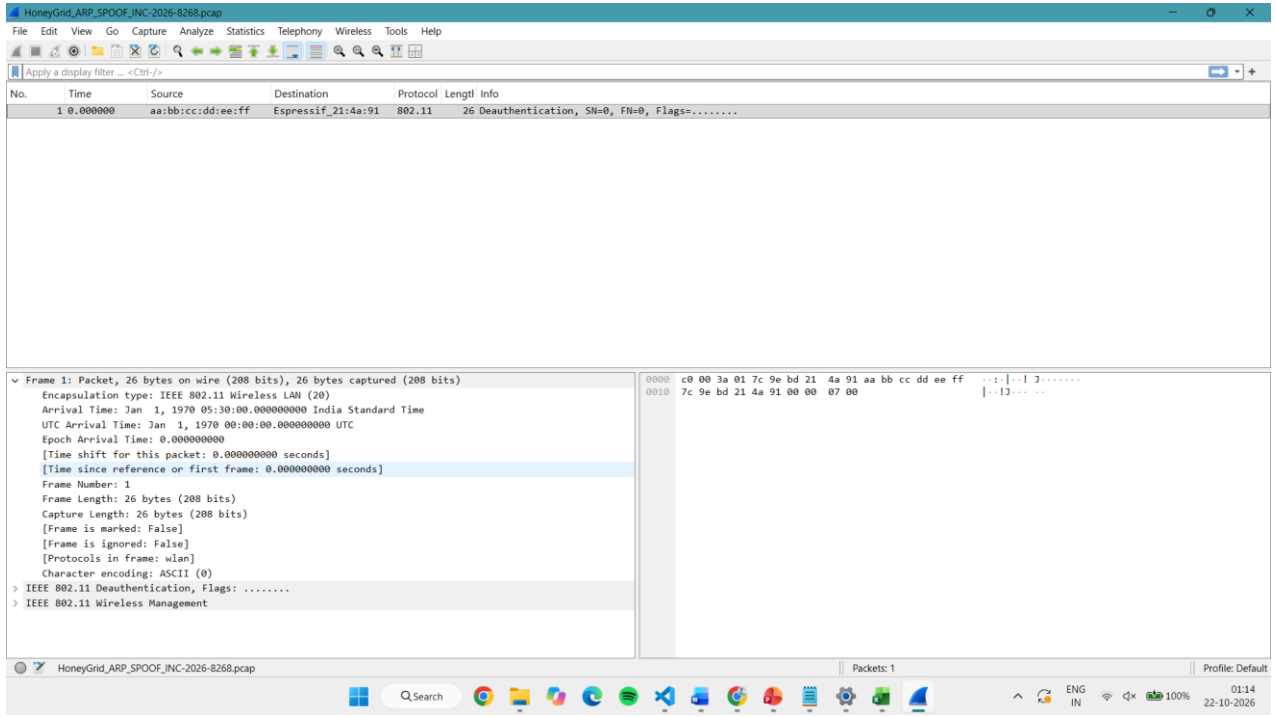

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:20:27 IST
JM
Live 802.11 & Network Frame Sniffer
Pause Sniffer
Clear Stream
18:50:27 [HARDWARE] ARP_SPOOF Packet detected from 192.168.4.21
18:50:25 [HARDWARE] PORT_SCAN Packet detected from 192.168.4.110
18:50:23 [HARDWARE] HTTP_FLOOD Packet detected from 192.168.4.88
18:50:21 [HARDWARE] BRUTE_FORCE Packet detected from 192.168.4.105
18:50:19 [HARDWARE] KARMA Packet detected from 12:34:56:78:90:AB
18:50:17 [HARDWARE] EVIL_TWIN Packet detected from 7C:9E:BD:21:4A:91
18:50:15 [HARDWARE] ARP_SPOOF Packet detected from 192.168.4.21
18:50:13 [HARDWARE] PORT_SCAN Packet detected from 192.168.4.110
18:50:11 [HARDWARE] HTTP_FLOOD Packet detected from 192.168.4.88
18:50:09 [HARDWARE] BRUTE_FORCE Packet detected from 192.168.4.105
18:50:07 [HARDWARE] KARMA Packet detected from 12:34:56:78:90:AB
18:50:05 [HARDWARE] EVIL_TWIN Packet detected from 7C:9E:BD:21:4A:91
18:50:03 [HARDWARE] ARP_SPOOF Packet detected from 192.168.4.21
18:50:01 [HARDWARE] PORT_SCAN Packet detected from 192.168.4.110
18:49:59 [HARDWARE] HTTP_FLOOD Packet detected from 192.168.4.88
18:49:57 [HARDWARE] BRUTE_FORCE Packet detected from 192.168.4.105
18:49:55 [HARDWARE] KARMA Packet detected from 12:34:56:78:90:AB
18:49:53 [HARDWARE] EVIL_TWIN Packet detected from 7C:9E:BD:21:4A:91

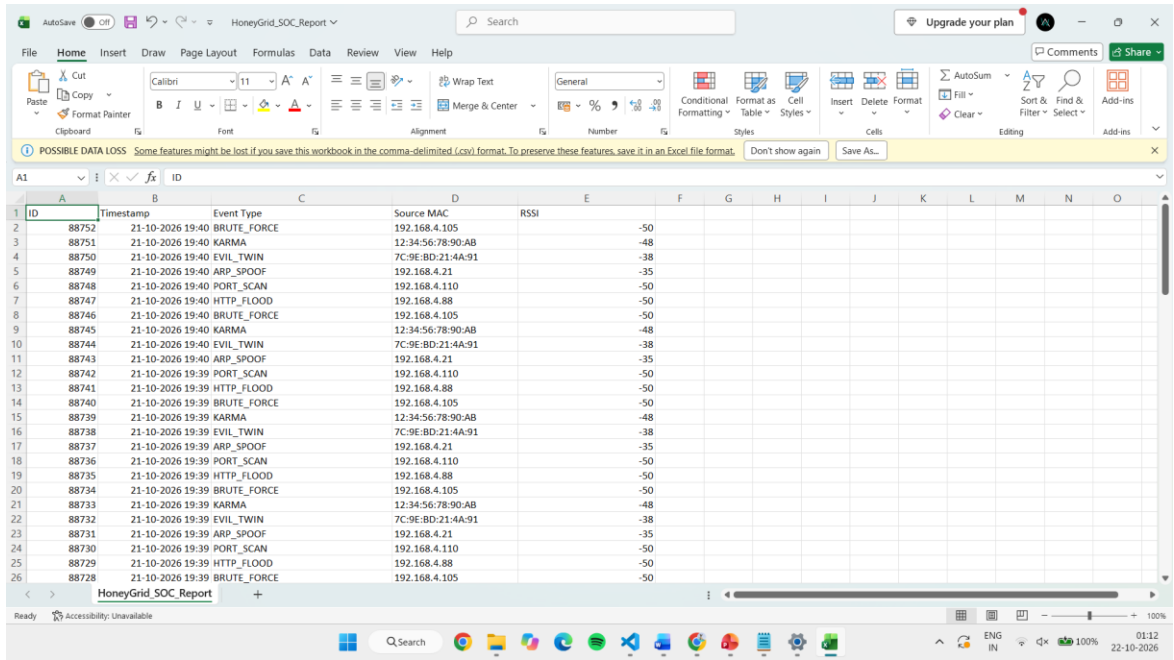

HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
02:05:58 IST
JM
Live 802.11 & Network Frame Sniffer
Pause Sniffer
Clear Stream
20:32:22 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR
20:32:19 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR
20:32:18 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR
20:26:58 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:56 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:53 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:51 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:49 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:48 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:46 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:44 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:42 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:40 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:39 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:35 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:33 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:32 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:30 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR


HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
00:20:52 IST
JM
SOC Analytics & Executive Compliance Reports
Select date range and format to export security assessment logs.
01 - 09 - 2026
26 - 09 - 2026
Generate PDF Report
Export CSV
HoneyGrid Wireless & Network Incident Report
Timestamp
Event Type
Source MAC/IP
Target BSSID/Node
RSSI / Status
2026-10-21 13:20:51
ARP_SPOOF
192.168.4.21
Campus_WiFi Node
-35 dBm
2026-10-21 13:20:49
PORT_SCAN
192.168.4.110
Campus_WiFi Node
-50 dBm
2026-10-21 13:20:47
HTTP_FLOOD
192.168.4.88
Campus_WiFi Node
-50 dBm
2026-10-21 13:20:45
BRUTE_FORCE
192.168.4.105
Campus_WiFi Node
-50 dBm
2026-10-21 13:20:43
KARMA
12:34:56:78:90:AB
Campus_WiFi Node
-48 dBm
2026-10-21 13:20:41
EVIL_TWIN
7C:9E:BD:21:4A:91
Campus_WiFi Node
-38 dBm
2026-10-21 13:20:39
ARP_SPOOF
192.168.4.21
Campus_WiFi Node
-35 dBm
2026-10-21 13:20:37
PORT_SCAN
192.168.4.110
Campus_WiFi Node
-50 dBm
2026-10-21 13:20:35
HTTP_FLOOD
192.168.4.88
Campus_WiFi Node
-50 dBm
2026-10-21 13:20:33
BRUTE_FORCE
192.168.4.105
Campus_WiFi Node
-50 dBm

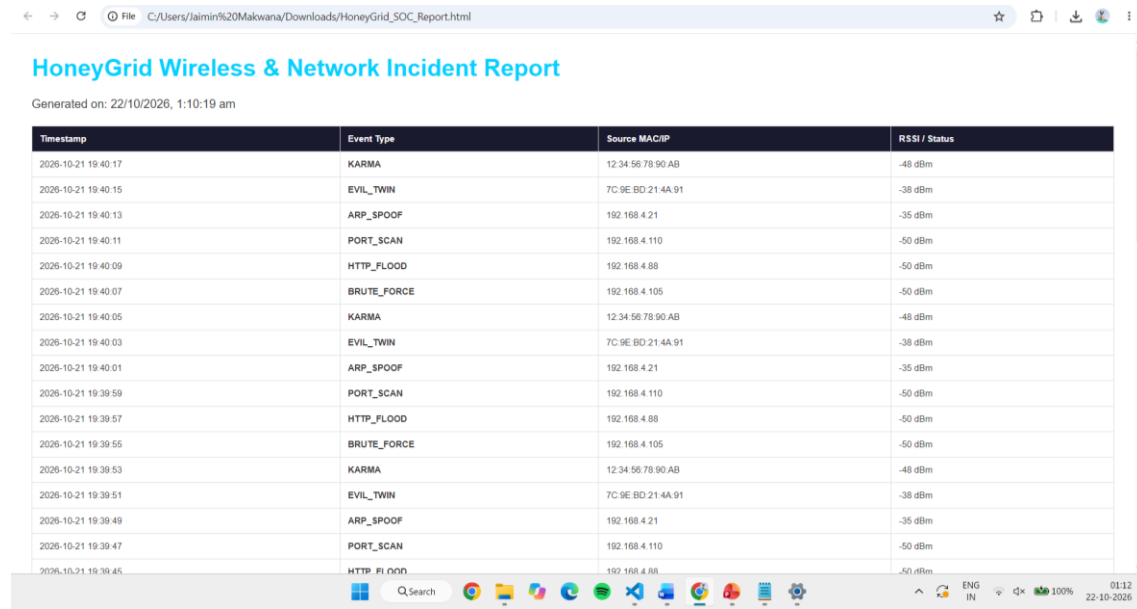
Also, we generate PDF Report, Export CSV and Download Case PCAP







ID	Timestamp	Event Type	Source MAC	RSSI
88752	21-10-2026 19:40	BRUTE_FORCE	192.168.4.105	-50
88751	21-10-2026 19:40	KARMA	12:34:56:78:90:AB	-48
88750	21-10-2026 19:40	EVIL_TWIN	7C:9E:BD:21:4A:91	-38
88749	21-10-2026 19:40	ARP_SPOOF	192.168.4.21	-35
88748	21-10-2026 19:40	PORT_SCAN	192.168.4.110	-50
88747	21-10-2026 19:40	HTTP_FLOOD	192.168.4.88	-50
88746	21-10-2026 19:40	BRUTE_FORCE	192.168.4.105	-50
88745	21-10-2026 19:40	KARMA	12:34:56:78:90:AB	-48
88744	21-10-2026 19:40	EVIL_TWIN	7C:9E:BD:21:4A:91	-38
88743	21-10-2026 19:40	ARP_SPOOF	192.168.4.21	-35
88742	21-10-2026 19:39	PORT_SCAN	192.168.4.110	-50
88741	21-10-2026 19:39	HTTP_FLOOD	192.168.4.88	-50
88740	21-10-2026 19:39	BRUTE_FORCE	192.168.4.105	-50
88739	21-10-2026 19:39	KARMA	12:34:56:78:90:AB	-48
88738	21-10-2026 19:39	EVIL_TWIN	7C:9E:BD:21:4A:91	-38
88737	21-10-2026 19:39	ARP_SPOOF	192.168.4.21	-35
88736	21-10-2026 19:39	PORT_SCAN	192.168.4.110	-50
88735	21-10-2026 19:39	HTTP_FLOOD	192.168.4.88	-50
88734	21-10-2026 19:39	BRUTE_FORCE	192.168.4.105	-50
88733	21-10-2026 19:39	KARMA	12:34:56:78:90:AB	-48
88732	21-10-2026 19:39	EVIL_TWIN	7C:9E:BD:21:4A:91	-38
88731	21-10-2026 19:39	ARP_SPOOF	192.168.4.21	-35
88730	21-10-2026 19:39	PORT_SCAN	192.168.4.110	-50
88729	21-10-2026 19:39	HTTP_FLOOD	192.168.4.88	-50
88728	21-10-2026 19:39	BRUTE_FORCE	192.168.4.105	-50



HoneyGrid Wireless & Network Incident Report

Generated on: 22/10/2026, 1:10:19 am

Timestamp	Event Type	Source MAC/IP	RSSI / Status
2026-10-21 19:40:17	KARMA	12:34:56:78:90:AB	-48 dBm
2026-10-21 19:40:15	EVIL_TWIN	7C:9E:BD:21:4A:91	-38 dBm
2026-10-21 19:40:13	ARP_SPOOF	192.168.4.21	-35 dBm
2026-10-21 19:40:11	PORT_SCAN	192.168.4.110	-50 dBm
2026-10-21 19:40:09	HTTP_FLOOD	192.168.4.88	-50 dBm
2026-10-21 19:40:07	BRUTE_FORCE	192.168.4.105	-50 dBm
2026-10-21 19:40:05	KARMA	12:34:56:78:90:AB	-48 dBm
2026-10-21 19:40:03	EVIL_TWIN	7C:9E:BD:21:4A:91	-38 dBm
2026-10-21 19:40:01	ARP_SPOOF	192.168.4.21	-35 dBm
2026-10-21 19:39:59	PORT_SCAN	192.168.4.110	-50 dBm
2026-10-21 19:39:57	HTTP_FLOOD	192.168.4.88	-50 dBm
2026-10-21 19:39:55	BRUTE_FORCE	192.168.4.105	-50 dBm
2026-10-21 19:39:53	KARMA	12:34:56:78:90:AB	-48 dBm
2026-10-21 19:39:51	EVIL_TWIN	7C:9E:BD:21:4A:91	-38 dBm
2026-10-21 19:39:49	ARP_SPOOF	192.168.4.21	-35 dBm
2026-10-21 19:39:47	PORT_SCAN	192.168.4.110	-50 dBm
2026-10-21 19:39:45	HTTP_FLOOD	192.168.4.88	-50 dBm

ESP32 Promiscuous Sniffer Firmware Code




```
#include <WiFi.h>
#include "esp_wifi.h"
#include "nvs_flash.h"
#include "esp_netif.h"

// DEAUTHER.NET SPECIFIC BURST THRESHOLDS (Window = 1.2 Seconds)
#define EVAL_WINDOW_MS 1200
#define DEAUTH_FLOOD_LIMIT 1    // Normal Deauth flood send rates
#define BEACON_FLOOD_LIMIT 5    // High threshold to filter real router beacons (~10/s)
#define PROBE_FLOOD_LIMIT 2     // High threshold to filter normal mobile probes

struct MacTracker {
  String mac;
  int deauthCount;
  int beaconCount;
  int probeCount;
  unsigned long windowStart;
};

#define MAX_TRACKERS 50
MacTracker trackers[MAX_TRACKERS];

String macToString(const uint8_t* mac) {
  char macStr[18];
  snprintf(macStr, sizeof(macStr), "%02X:%02X:%02X:%02X:%02X:%02X",
           mac[0], mac[1], mac[2], mac[3], mac[4], mac[5]);
  return String(macStr);
}

void processPacket(uint8_t type, String mac, int8_t rssi) {
  unsigned long now = millis();
  int idx = -1;

  for (int i = 0; i < MAX_TRACKERS; i++) {
    if (trackers[i].mac == mac) {
      idx = i;
      break;
    }
  }

  if (idx == -1) {
    for (int i = 0; i < MAX_TRACKERS; i++) {
      if (trackers[i].mac == "" || (now - trackers[i].windowStart > EVAL_WINDOW_MS)) {
```

How to connect Esp32 on Arduino

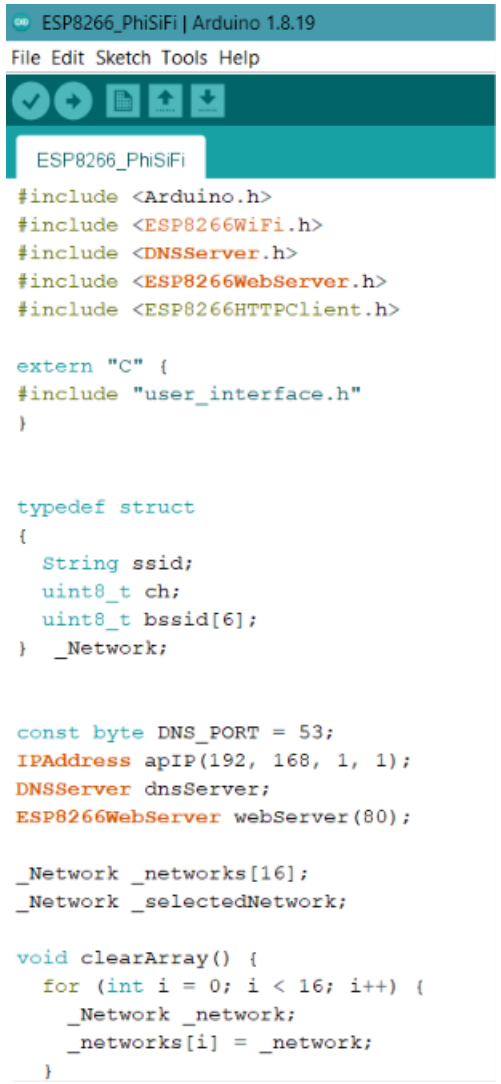
Step1: open Preference and paste this Link : https://raw.githubusercontent.com/espressif/arduino-esp32/gh-pages/package_esp32_index.json
Install Esp32 Package

Step2: Go to Boards and Search your Boards name (select Esp32 Dev module)

Step 3: Upload speed (115200) (make sure same speed on Sever.js code and also ESP8266 attacker code)

Step4: Select port (make sure select laptop port number) (COM7)
Step5: Flask code in ESP32

ESP8266 Attack code




```
#include <Arduino.h>
#include <ESP8266WiFi.h>
#include <DNSServer.h>
#include <ESP8266WebServer.h>
#include <ESP8266HTTPClient.h>

extern "C" {
#include "user_interface.h"
}


typedef struct
{
  String ssid;
  uint8_t ch;
  uint8_t bssid[6];
}  _Network;


const byte DNS_PORT = 53;
IPAddress apIP(192, 168, 1, 1);
DNSServer dnsServer;
ESP8266WebServer webServer(80);

_Network _networks[16];
_Network _selectedNetwork;

void clearArray() {
  for (int i = 0; i < 16; i++) {
    _Network _network;
    _networks[i] = _network;
  }
```

How to connect Esp8266 on Arduino

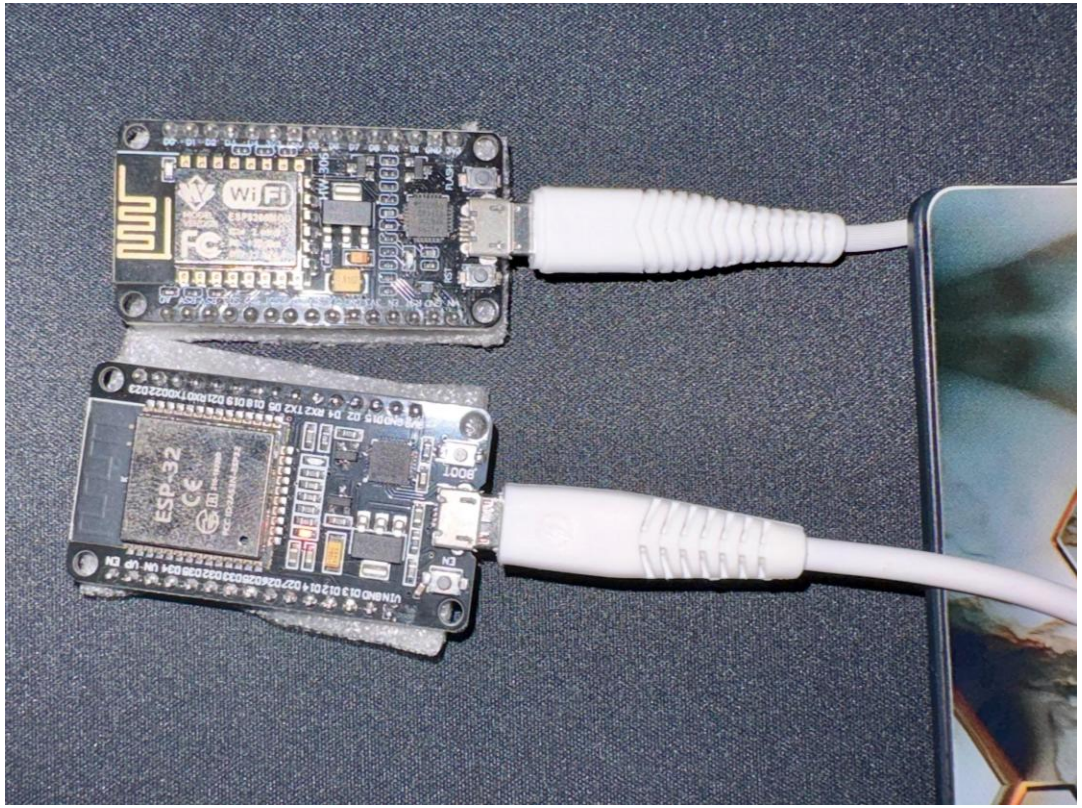
Step1: open Preference and paste this Link :
http://arduino.esp8266.com/stable/package_esp8266com_index.json
Install Esp8266 Package

Step2: Go to Boards and Search your Boards name (select Generic Esp8266 Nodemcu)

Step 3: Upload speed (115200) (make sure same speed on Sever.js code and also ESP8266 attacker code)

Step4: Select port (make sure select laptop port number) (COM7)
Step5: Flask code in ESP8266

Physical Hardware Interfacing & Components View

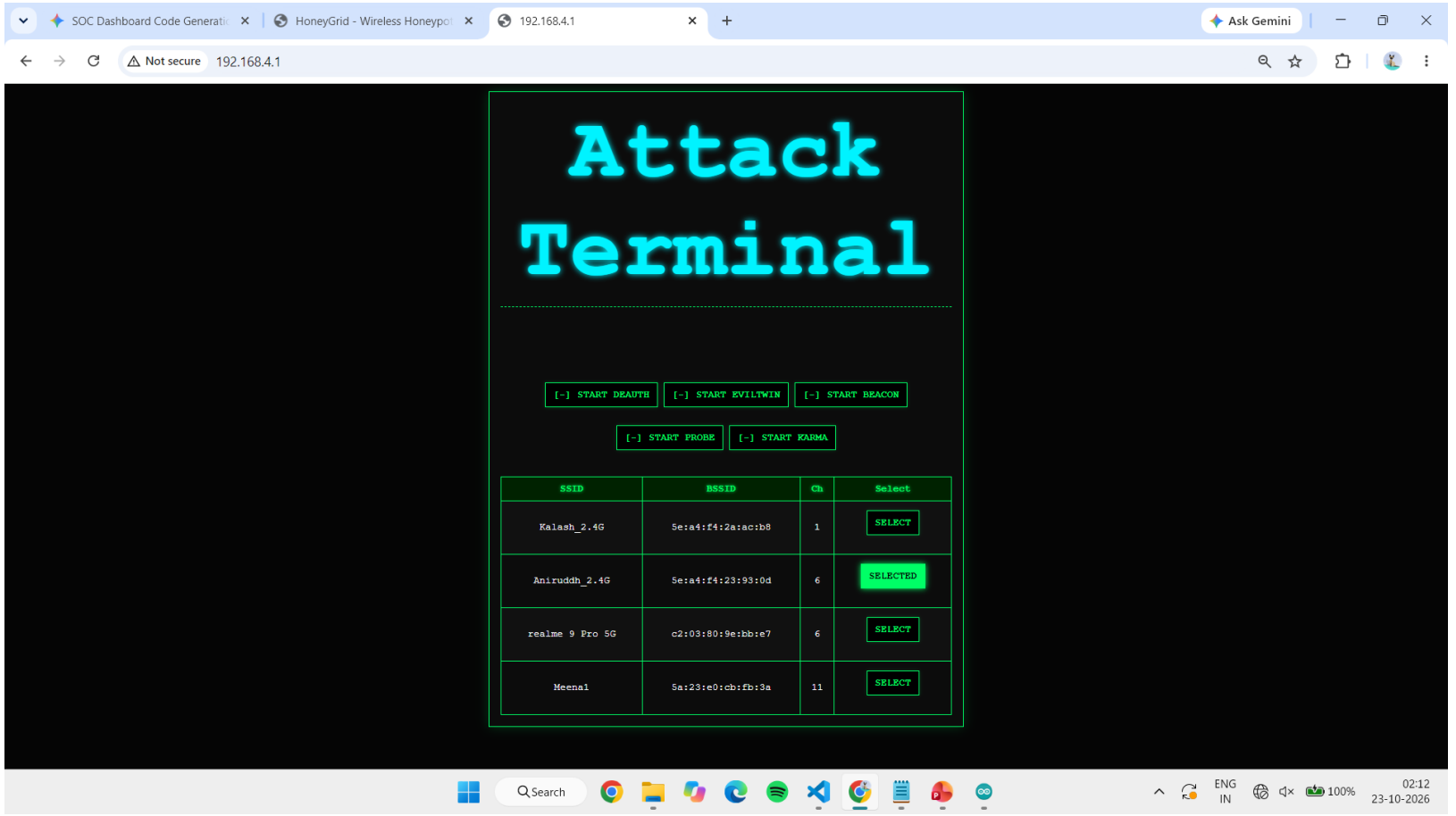


ESP32 NODEMCU USING ATTACK DETECTOR
ESP8266 NODEMCU USED BY ATTACKER

ESP8266 Deauther web panel

Step 1: Connect ESP8266 nodemcu any powered device.
Step 2: Turn On WIFI and connect ESP8266 WIFI (wife name : HACKER)
Step 3: Open Browser, Search 192.168.4.1
Step 4: Scan APS
Step 5: Select Access points
Step 6: Start Attack (Deauth, Probe, Karma, Beacon and evil-twin attacks)

Execute Web panel



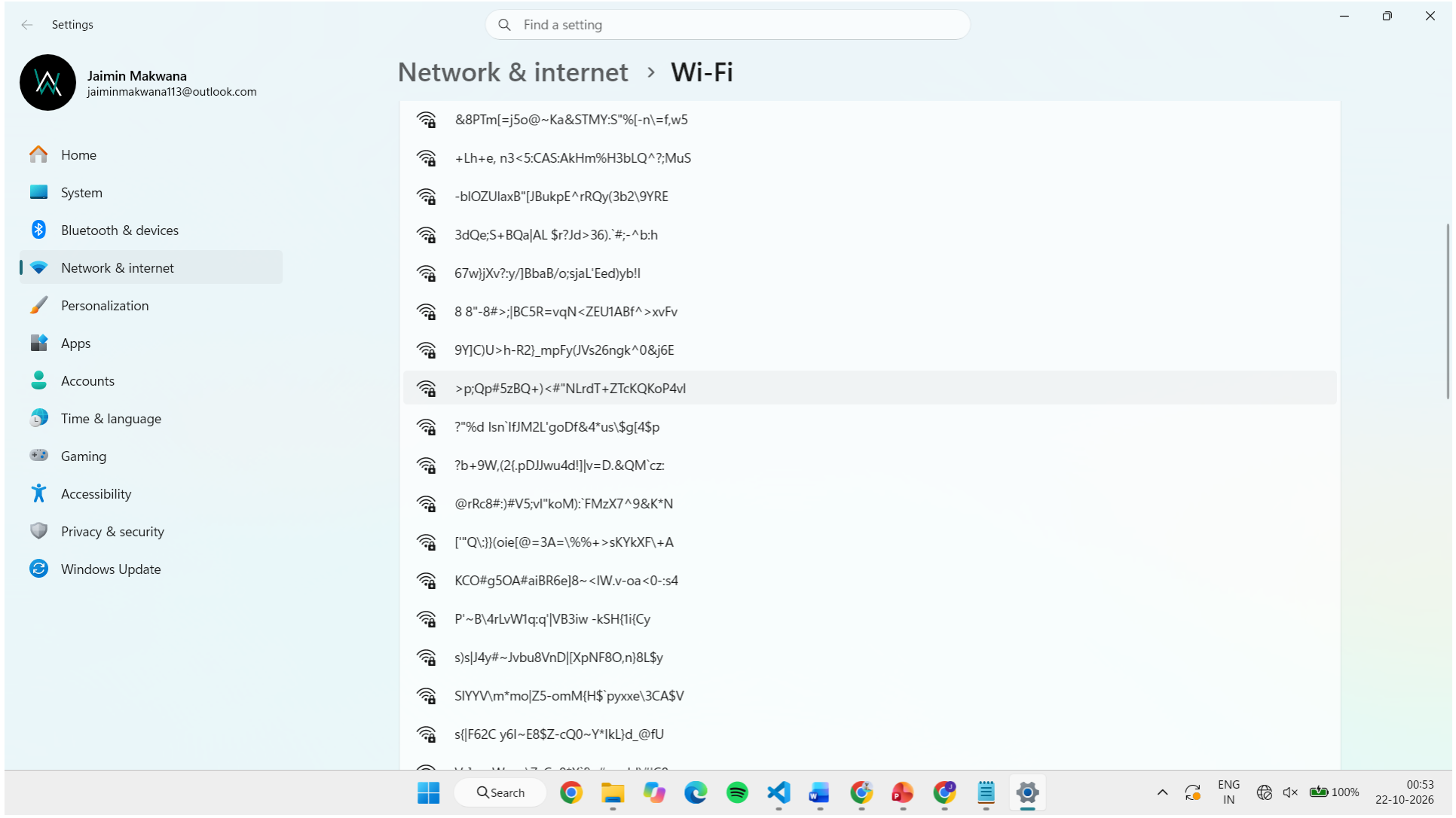
Attack Terminal

[-] START DEAUTH | [-] START EVILTWIN | [-] START BEACON

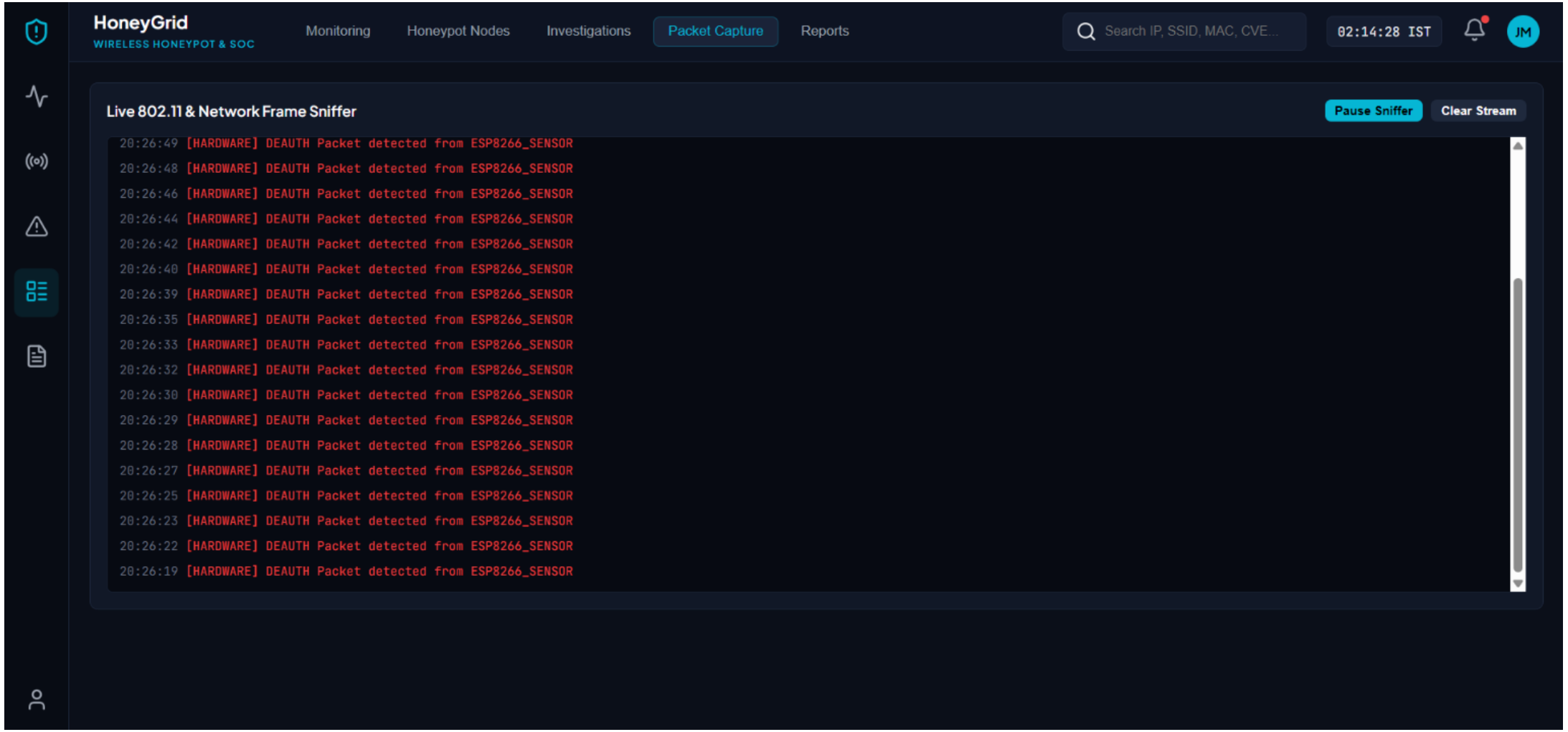
[-] START PROBE | [-] START KARMA

SSID	BSSID	Ch	Select
Kalash_2.4G	5e:a4:f4:2a:ac:b8	1	SELECT
Aniruddh_2.4G	5e:a4:f4:23:93:0d	6	SELECTED
realme 9 Pro 5G	c2:03:80:9e:bb:e7	6	SELECT
Meena1	5a:23:e0:cb:fb:3a	11	SELECT

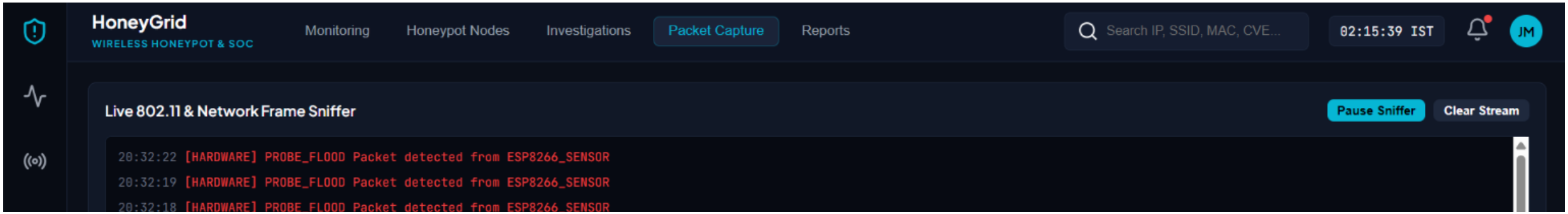
Beacon flood is successfully run



Deauth


HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
02:14:28 IST
JM
Live 802.11 & Network Frame Sniffer
Pause Sniffer
Clear Stream
20:26:49 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:48 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:46 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:44 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:42 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:40 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:39 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:35 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:33 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:32 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:30 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:29 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:28 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:27 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:25 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:23 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:22 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR
20:26:19 [HARDWARE] DEAUTH Packet detected from ESP8266_SENSOR

Probe flood


HoneyGrid
WIRELESS HONEYPOT & SOC
Monitoring
Honeypot Nodes
Investigations
Packet Capture
Reports
Search IP, SSID, MAC, CVE...
02:15:39 IST
JM
Live 802.11 & Network Frame Sniffer
Pause Sniffer
Clear Stream
20:32:22 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR
20:32:19 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR
20:32:18 [HARDWARE] PROBE_FLOOD Packet detected from ESP8266_SENSOR

Open Database
http://localhost:3000/api/logs



Conclusion

The Smart Wireless Honeypot and Real-Time Security Operations Center (SOC) Dashboard demonstrates a practical approach to wireless threat monitoring using low-cost embedded hardware and a web-based interface. The ESP32 sniffer observes selected 802.11 management-frame activity, while the ESP8266 is used as a honeypot or wireless test node. Serial communication connects the hardware path to a Node.js/Express backend, and the dashboard provides a centralized view of security events.

The project presentation covers nine threat vectors and includes physical demonstrations, software-generated test events, log access, and export functionality. Its modular design makes it useful for learning how embedded monitoring, backend processing, and SOC visualization can work together.

The system is intended for educational and authorized defensive testing. Further work can improve detection logic, event correlation, dashboard analysis, and the reliability of logging and reporting.

Thank You !!